Undirected Graphical Models: Learning Structure from Data

DS6410 Machine Learning II

What We Will Cover Today

This lecture progresses through five interconnected themes, building from motivation to methodology to application.

#	Topic	*Key Question*
1	Graphical Models in Unsupervised Learning	*Why do we need them?*
2	Markov Random Fields & Cliques	*How do we represent structure?*
3	Marginal vs. Conditional Independence	*What does an edge really mean?*
4	Gaussian Graphical Models	*How do we estimate the graph?*
5	Extensions & Applications	*Where does this go in practice?*

Closing: Preview of Directed Graphical Models (next lecture)

Representing Joint Distributions in High Dimensions Is Intractable

In unsupervised learning, we observe data drawn from an unknown joint distribution. Our goal is to learn something meaningful about this distribution — its structure, its dependencies, and its conditional independencies.

The fundamental challenge is the **curse of dimensionality**. As the number of variables grows, the number of parameters required to specify the full joint distribution grows exponentially (for discrete variables) or quadratically (for continuous variables).

THE SOLUTION

Exploit **conditional independence structure**. If many pairs of variables are conditionally independent given others, we can represent the joint distribution compactly using a graph, drastically reducing the number of parameters to estimate.

Data Observation

$$\mathbf{X} = (X_1, X_2, \ldots, X_p) \sim P_X$$

Binary Variables

$$2^p - 1 \text{ free parameters}$$

For $p = 100$, this is astronomically large.

Continuous Variables

$$O(p^2) \text{ parameters}$$

Requires $n > p$ observations for invertibility of the covariance matrix.

Graphical Models Encode Conditional Independence as Graph Structure

A **probabilistic graphical model** uses a graph-based representation as the foundation for encoding a complete distribution over a multi-dimensional space. There are two broad families of graphical models:

Type	Graph	Edges	Encodes
Undirected (Markov Random Field)	Undirected	Symmetric associations	Soft constraints, compatibility
Directed (Bayesian Network)	DAG	Causal arrows	Generative process, causality

> **Today's Focus: Undirected graphical models — ideal for learning associations without assuming directionality.**

The Graph

$$G = (V, E)$$

Represents a family of joint distributions over variables $\{X_1, \ldots, X_p\}$.

Nodes

$$v \in V$$

Each node corresponds to a random variable X_v.

Edges

$$(u, v) \in E$$

Each edge (or absence of an edge) encodes a specific conditional independence statement.

UNSUPERVISED TASKS

- Gene regulatory network discovery
- Brain functional connectivity mapping
- Social network analysis
- Topic modeling

Graphical Models Are a Core Tool for Unsupervised Structure Discovery

In unsupervised learning, we have no labels. Our goal is to discover the latent structure in the joint distribution P_X. Graphical models serve this goal in several fundamental ways:

- **Density estimation:** The graph defines a factored representation of P_X, enabling tractable estimation even in high dimensions.
- **Structure learning:** We learn the graph topology directly from data to answer: which variables are directly connected?
- **Exploratory analysis:** The learned graph reveals clusters, hubs, and pathways in complex datasets.
- **Downstream inference:** Once learned, the graph supports probabilistic queries such as $P(X_i \mid X_j = x)$ and finding most probable configurations.

Joint Distribution Factorization

$$P(X_1, \ldots, X_p) = \frac{1}{Z} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{X}_c)$$

Partition Function

$$Z = \sum_{\mathbf{x}} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{x}_c)$$

The normalizing constant ensuring the distribution sums to 1.

A Markov Random Field Is an Undirected Graph with Probability Potentials

Definition: A **Markov Random Field (MRF)**, also called a Markov network or undirected graphical model (UGM), is a pair (G, Ψ) where:

- $G = (V, E)$ is an undirected graph.
- $\Psi = \{\psi_c : c \in \mathcal{C}\}$ is a set of **potential functions** (also called factors), one per clique c in G.

KEY POINT

Potential functions $\psi_c \geq 0$ are **not probabilities** — they encode compatibility or affinity between variable configurations in clique c.

Factorization over Maximal Cliques

$P(X) = (1 / Z) \cdot \psi_{123}(X_1,X_2,X_3) \cdot \psi_{234}(X_2,X_3,X_4) \cdot \psi_{45}(X_4,X_5)$

Pairwise MRFs

In practice, we typically use pairwise MRFs where potentials are defined over edges (cliques of size 2).

The Graph's Clique Structure Determines the Factorization

Definition: A **clique** in graph G is a fully connected subset of nodes — every pair of nodes in the clique is connected by an edge. A **maximal clique** is a clique that cannot be extended by adding another node.

EXAMPLE GRAPH STRUCTURE

- **Nodes:** {1, 2, 3, 4, 5}
- **Edges:** (1,2), (1,3), (2,3), (2,4), (3,4), (4,5)
- **Cliques:** {1,2}, {1,3}, {2,3}, {1,2,3}, {2,4}, {3,4}, {2,3,4}, {4,5}
- **Maximal Cliques:** {1,2,3}, {2,3,4}, {4,5}

The joint distribution factorizes as a product of potential functions, one for each maximal clique in the graph. This drastically reduces the complexity of representing the distribution.

Gibbs Distribution

$$P(\mathbf{X}|\theta) = \frac{1}{Z(\theta)} \exp\left(-\sum_{c \in \mathcal{C}} E_c(\mathbf{X}_c \mid \theta_c) \right)$$

Energy Function

$E_c(\mathbf{X}_c)$ is an **energy function** for clique c.

Lower energy corresponds to higher probability (more compatible configurations).

The Gibbs Distribution Connects MRFs to Energy-Based Models

The **Gibbs distribution** (also known as the Boltzmann distribution in statistical physics) provides an alternative way to write the joint distribution of a Markov Random Field using energy functions instead of potential functions.

> **HAMMERSLEY-CLIFFORD THEOREM**
>
> Any strictly positive distribution $P(\mathbf{X}) > 0$ satisfies the Markov properties of graph G **if and only if** it can be written as a product of clique potentials (i.e., as a Gibbs distribution over G).

This theorem is foundational: it guarantees that the **graphical structure** (conditional independencies) and the **probabilistic structure** (factorization over cliques) are mathematically equivalent for positive distributions.

Three Equivalent Markov Properties Define the Independence Structure

For a graph $G = (V, E)$ and distribution P, three Markov properties are defined to formalize conditional independence:

Property	Statement
Global	$X_A \perp\!\!\!\perp X_B \mid X_C$ whenever C separates A from B in G
Local	$X_s \perp\!\!\!\perp X_{V \setminus \{s\} \setminus N(s)} \mid X_{N(s)}$ A node is independent of non-neighbors given its neighbors.
Pairwise	$X_s \perp\!\!\!\perp X_t \mid X_{V \setminus \{s,t\}}$ For every non-adjacent pair (s, t).

Relationship

Global $\Rightarrow$ Local $\Rightarrow$ Pairwise

> **Under Positivity:**
> All three are equivalent (Hammersley-Clifford).

Markov Blanket

The Markov blanket of node s is its set of neighbors $N(s)$.

Given $N(s)$, node s is independent of all other nodes in the graph.

What Pearson Correlation Sees

The Consequence

A correlation-based network draws edges between all pairs, creating **spurious edges** where no direct relationship exists.

Pearson Correlation Measures Marginal Association — and That Is Not Enough

Setup: Three variables: Z (exercise), X (diet), and Y (weight loss).

True Process: Z and X both cause Y, and Z also influences X (people who exercise tend to eat better).

Because the variables are interconnected, they will all be marginally correlated. A network built on Pearson correlation will suggest direct links everywhere and cannot distinguish direct from indirect or common-cause relationships.

THE DEEPER PROBLEM

Pearson correlation is a **marginal measure**. It does not condition on other variables, so indirect pathways or common causes can produce correlations that are not direct relationships.

Marginal Correlation Creates Spurious Edges in a Simple Three-Variable System

The True Structure: X and Y are conditionally independent given Z — they are linked only through the common cause Z.

The Illusion of Marginal Correlation: Marginal Cor(X,Y) is positive; with small σ^2 it can be large, producing a false edge in correlation-based networks.

The Reality of Partial Correlation: Conditioning on Z yields zero partial correlation, revealing no direct X–Y link.

> **CONCLUSION**
>
> Pearson-correlation networks overstate connectivity and can introduce false edges when common causes or mediators exist.

Example Setup

$$Z \sim \mathcal{N}(0,1)$$

$$X = Z + \varepsilon_X, \quad Y = Z + \varepsilon_Y$$

$$\varepsilon_X, \varepsilon_Y \overset{iid}{\sim} \mathcal{N}(0,\sigma^2)$$

Marginal Correlation

$$\mathrm{Cor}(X,Y) = \frac{\mathrm{Var}(Z)}{\mathrm{Var}(Z)+\sigma^2} > 0$$

Partial Correlation

$$\mathrm{Cor}(X,Y \mid Z) = 0$$

Precision Matrix

$$\Theta = \Sigma^{-1}$$

Partial Correlation

For Gaussian data, partial correlation has a closed form via the precision matrix:

$$\rho_{ij|\text{rest}} = -\frac{\Theta_{ij}}{\sqrt{\Theta_{ii}\Theta_{jj}}}$$

Partial Correlation Measures Direct Association After Removing Shared Effects

Definition: The **partial correlation** between X_i and X_j given all other variables $X_{V\setminus\{i,j\}}$ is the correlation of the residuals after regressing X_i and X_j on all other variables.

KEY INSIGHT

The **sparsity pattern of the precision matrix** is exactly the **edge set of the conditional independence graph**.

Measure	Conditions on	Detects
Pearson correlation	Nothing	Marginal association (direct + indirect)
Partial correlation	All other variables	Direct conditional dependence only

Precision Matrix

$$\Theta = \Sigma^{-1}$$

Edge Condition

$$\text{Edge } (i,j) \in E \iff \Theta_{ij} \neq 0$$

Joint Density

$$P(\mathbf{X}) \propto \exp\left(-\tfrac{1}{2}\mathbf{X}^\top \Theta \mathbf{X}\right)$$

Gaussian Graphical Model & Precision Matrix

Definition: A Gaussian Graphical Model (GGM) is an undirected model where

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

and the graph encodes conditional independence via

$$\Theta = \Sigma^{-1}.$$

Why GGMs are powerful:

- Conditional independence readable from entries of Θ.
- Edges represent direct partial correlations.
- Learning reduces to estimating a sparse precision matrix.

LIMITATION

Standard GGM assumes multivariate Gaussianity; nonparanormal transformations are a common fix.

Log-Likelihood

$$\ell(\Theta) = \frac{n}{2}[\log\det(\Theta) - \mathrm{tr}(S\Theta)]$$

Sample Covariance

$$S = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i^\top$$

Maximum Likelihood Estimate

$$\hat{\Theta}_{MLE} = S^{-1}$$

Requires S to be invertible, which requires $n > p$.

The MLE of the Precision Matrix Fails When $p \geq n$

Given n i.i.d. observations $\mathbf{x}_1, \ldots, \mathbf{x}_n \sim \mathcal{N}(\mathbf{0}, \Sigma)$, the standard approach is to find the Maximum Likelihood Estimate (MLE) of the precision matrix Θ.

The High-Dimensional Problem: In modern applications, the number of variables p is often much larger than the number of samples n ($p \gg n$):

- **Genomics:** $p = 20,000$ genes, $n = 200$ samples
- **Neuroimaging:** $p = 1,000$ brain regions, $n = 100$ subjects
- **Result:** The sample covariance matrix S is singular (not invertible), so the MLE does not exist.

THE SOLUTION

Regularization. We must impose structural assumptions on Θ, specifically **sparsity**, to make estimation possible and to recover a meaningful, interpretable graph.

Neighborhood Selection Estimates the Graph via Node-Wise LASSO Regressions

Node-Wise LASSO

For each node $i = 1, \ldots, p$:

$$\hat{\beta}^{(i)} = \arg\min_{\beta} \left\{ \frac{1}{n} \| X_i - X_{-i}\beta \|_2^2 + \lambda \| \beta \|_1 \right\}$$

Edge Inclusion Rule

Edge (i, j) is included if:

OR Rule:

$$\hat{\beta}_j^{(i)} \neq 0 \quad \text{or} \quad \hat{\beta}_i^{(j)} \neq 0$$

AND Rule:

$$\hat{\beta}_j^{(i)} \neq 0 \quad \text{and} \quad \hat{\beta}_i^{(j)} \neq 0$$

KEY INSIGHT (MEINSHAUSEN & BÜHLMANN, 2006)

In a Gaussian Graphical Model, conditional independence between variables corresponds to zero regression coefficients when regressing one variable on the others.

Estimate each node's neighborhood independently via LASSO to induce sparsity.

Properties of Neighborhood Selection:

- **Computationally efficient:**
- **Consistent:**
- **Structure only:**

The Graphical LASSO Directly Estimates a Sparse Precision Matrix

Graphical LASSO (Friedman, Hastie & Tibshirani, 2008) maximizes the ℓ_1-penalized log-likelihood to directly estimate a sparse precision matrix.

ALGORITHM & PERFORMANCE

Algorithm: Uses block coordinate descent — solves a sequence of LASSO problems on rows/columns of Θ.

Practical Performance: Highly efficient. It can solve a $p = 1,000$ node problem in under 1 minute.

Tuning the Regularization Parameter (λ):

- **Large λ:** Sparser graph (fewer edges), higher bias.
- **Small λ:** Denser graph, lower bias but higher variance.
- **Selection Methods:** BIC, cross-validation, or stability selection (StARS).

Graphical LASSO Objective

$$\hat{\Theta} = \arg\max_{\Theta \succ 0} \{\log\det(\Theta) - \mathrm{tr}(S\Theta) - \lambda\|\Theta\|_{1,\mathrm{off}}\}$$

L1 Penalty

$$\|\Theta\|_{1,\mathrm{off}} = \sum_{i \neq j} |\Theta_{ij}|$$

Sparsity

The true graph has maximum degree , where .

Irrepresentability Condition

Variables outside the true neighborhood cannot be too correlated with variables inside it.

Sample Size Requirement

Samples must grow roughly as:

Consistent Graph Recovery Is Possible Under Sparsity and Regularity Conditions

Consistent Graph Recovery: As , the probability of recovering the exact edge set approaches 1.

THEORETICAL IMPLICATIONS

Even when , recovery is possible provided the graph is sparse and .

Why this matters:

- **High-Dimensional Validity:** Justifies Graphical LASSO in genomics and neuroimaging.
- **Interpretability:** Guarantees edges reflect direct associations, not artifacts.

Modern GGM Extensions Handle Non-Gaussian Data, Multiple Graphs, and Temporal Dynamics

Standard GGMs assume i.i.d. Gaussian data. Modern extensions relax this to handle real-world datasets efficiently.

- **Nonparanormal (Copula) GGMs:** Use monotonic transforms and rank correlations for robust graph estimates.
- **Joint Estimation (Multiple Graphs):** Estimate related graphs together to borrow strength across groups.
- **Time-Varying GGMs:** Track smoothly evolving graph structure over time.

> **KEY TAKEAWAY**
>
> GGM tools are flexible: change the input correlation or objective to adapt to different data and structure.

Nonparanormal Model

Assume monotonic transforms f_j so transformed marginals are Gaussian:

$$(f_1(X_1), \ldots, f_p(X_p)) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Joint Estimation (Fused GLASSO)

Estimate K related graphs jointly with penalties on sparsity and differences:

$$\sum_{k=1}^{K} \ell(\Theta^{(k)}) - \lambda_1 \sum_{k=1}^{K} \|\Theta^{(k)}\|_1 - \lambda_2 \sum_{k<k'} \|\Theta^{(k)} - \Theta^{(k')}\|_1$$

Belief Propagation Enables Efficient Marginal Inference on Graphical Models

After learning a model, we often want to compute probabilities such as $P(X_i \mid E)$ for queries given evidence.

- **Sum-Product Algorithm:** Belief Propagation (BP) computes marginals efficiently by passing messages along edges.
- **Exact on Trees:** For tree-structured graphs, BP converges to exact marginals in a finite number of steps.

> **LOOPY BELIEF PROPAGATION**
>
> On graphs with cycles, iterative message passing (LBP) often yields good approximate marginals despite no convergence guarantee.

Marginal Inference

Computing the marginal probability of a single node X_i:

$$P(X_i) = \sum_{X \setminus X_i} P(X)$$

Naively, this requires summing over exponentially many states.

Message Passing

Node j sends a message to node i:

$$m_{j \to i}(x_i) = \sum_{x_j} \psi(x_i, x_j)\psi(x_j) \prod_{k \in N(j) \setminus i} m_{k \to j}(x_j)$$

The Biological Problem

Marginal Co-expression: Genes often activate together due to shared regulation.

Pearson correlation then creates a dense, uninterpretable network.

The GGM Solution

Direct Interaction: Conditioning on other genes removes common-cause effects.

The resulting sparse graph reveals direct regulatory links.

GGMs Reveal Direct Gene-Gene Interactions Beyond Marginal Co-expression

One of the most successful applications of Gaussian Graphical Models is in **genomics** for discovering **Gene Regulatory Networks**.

How GGMs are used in Genomics:

- **Pathway Discovery:** Find conditionally dependent genes.
- **Hub Identification:** Detect master regulator genes.
- **Disease Mechanisms:** Compare healthy vs diseased net...

IMPACT

GGMs convert high-dimensional, correlated data into a sparse map of direct interactions to guide experiments.

GGMs Uncover Direct Connectivity in Neuroscience and Risk Structure in Finance

NEUROSCIENCE: BRAIN CONNECTIVITY

Data: fMRI BOLD signals across brain regions over time.

The Problem: Global physiological signals induce widespread correlations that obscure direct neural pathways.

> **The GGM Solution**
>
> Estimating a sparse precision matrix conditions out global signals and reveals the **direct functional connectivity** between regions, aiding disorder studies.

FINANCE: PORTFOLIO OPTIMIZATION

Data: Daily or high-frequency stock returns (e.g., S&P 500).

The Problem: The market factor creates near-universal correlations, complicating diversification.

> **The GGM Solution**
>
> GGMs remove the market factor to expose **sector-specific risk**; the precision matrix is also required for Markowitz optimization.

Undirected Graphical Models Provide a Principled Framework for Association Learning

Summary of Key Concepts from Lecture 1

1. CONDITIONAL INDEPENDENCE

Unlike marginal correlation (which creates spurious edges), **partial correlation** measures direct association. The graph structure encodes these conditional independencies via Global, Local, and Pairwise Markov properties.

2. FACTORIZATION & ENERGY

The joint probability distribution factorizes over the **maximal cliques** of the graph. By the Hammersley-Clifford theorem, this is equivalent to an energy-based **Gibbs distribution**.

3. GAUSSIAN GRAPHICAL MODELS

For multivariate Gaussian data, the conditional independence graph is encoded by the sparsity pattern of the **precision matrix** ($\Theta = \Sigma^{-1}$). Zero entries correspond to missing edges.

4. HIGH-DIMENSIONAL ESTIMATION

When $p \gg n$, the sample covariance is singular. We recover the graph using **regularization** techniques like Neighborhood Selection or the **Graphical LASSO**.

Next Lecture: When Direction Matters

Moving from symmetric associations to asymmetric causal relationships

TOPICS FOR LECTURE 2

- → **Directed Graphical Models:** Bayesian Networks and DAGs
- → **Conditional Independence:** D-Separation and the Explaining Away effect
- → **Factor Graphs:** Unifying directed and undirected models
- → **Exact Inference:** The Variable Elimination algorithm